



Chapter 5: Eigenvalues and Eigenvectors

Question 1:

1. Is $\lambda = 2$ an eigenvalue of $\begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$? Why or why not?
2. Is $\lambda = -2$ an eigenvalue of $\begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}$? Why or why not?
3. Is $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} -3 & 1 \\ -3 & 8 \end{bmatrix}$? If so, find the eigenvalue.
4. Is $\begin{bmatrix} -1 + \sqrt{2} \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$? If so, find the eigenvalue.
5. Is $\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$? If so, find the eigenvalue.

6. Is $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 3 & 6 & 7 \\ 3 & 3 & 7 \\ 5 & 6 & 5 \end{bmatrix}$? If so, find the eigenvalue.

7. Is $\lambda = 4$ an eigenvalue of $\begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix}$? If so, find one corresponding eigenvector.

8. Is $\lambda = 3$ an eigenvalue of $\begin{bmatrix} 1 & 2 & 2 \\ 3 & -2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$? If so, find one corresponding eigenvector.

Question 1: solution

1. The number 2 is an eigenvalue of A if and only if the equation $A\mathbf{x} = 2\mathbf{x}$ has a nontrivial solution. This equation is equivalent to $(A - 2I)\mathbf{x} = \mathbf{0}$. Compute

$$A - 2I = \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

The columns of A are obviously linearly dependent, so $(A - 2I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, and so 2 is an eigenvalue of A .

2. The number -2 is an eigenvalue of A if and only if the equation $A\mathbf{x} = -2\mathbf{x}$ has a nontrivial solution. This equation is equivalent to $(A + 2I)\mathbf{x} = \mathbf{0}$. Compute

$$A + 2I = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}$$

The columns of A are obviously linearly dependent, so $(A + 2I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, and so -2 is an eigenvalue of A .

3. Is $A\mathbf{x}$ a multiple of $\mathbf{x}$? Compute $\begin{bmatrix} -3 & 1 \\ -3 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 29 \end{bmatrix} \neq \lambda \begin{bmatrix} 1 \\ 4 \end{bmatrix}$. So $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ is *not* an eigenvector of A .

4. Is $A\mathbf{x}$ a multiple of $\mathbf{x}$? Compute $\begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} -1 + \sqrt{2} \\ 1 \end{bmatrix} = \begin{bmatrix} -1 + 2\sqrt{2} \\ 3 + \sqrt{2} \end{bmatrix}$ The second entries of $\mathbf{x}$ and $A\mathbf{x}$ shows

that if $A\mathbf{x}$ is a multiple of $\mathbf{x}$, then that multiple must be $3 + \sqrt{2}$. Check $3 + \sqrt{2}$ times the first entry of $\mathbf{x}$:

$$(3 + \sqrt{2})(-1 + \sqrt{2}) = -3 + (\sqrt{2})^2 + 2\sqrt{2} = -1 + 2\sqrt{2}$$

This matches the first entry of $A\mathbf{x}$, so $\begin{bmatrix} -1 + \sqrt{2} \\ 1 \end{bmatrix}$ is an eigenvector of A , and the corresponding eigenvalue is $3 + \sqrt{2}$.

5. Is $A\mathbf{x}$ a multiple of $\mathbf{x}$? Compute $\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. So $\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ is an eigenvector of A for the eigenvalue 0.

6. Is $A\mathbf{x}$ a multiple of $\mathbf{x}$? Compute $\begin{bmatrix} 3 & 6 & 7 \\ 3 & 3 & 7 \\ 5 & 6 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \\ -2 \end{bmatrix} = (-2) \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$. So $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ is an eigenvector of A for the eigenvalue -2 .

7. To determine if 4 is an eigenvalue of A , decide if the matrix $A - 4I$ is invertible.

$$A - 4I = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 0 & -1 \\ 2 & -1 & 1 \\ -3 & 4 & 1 \end{bmatrix}$$

Invertibility can be checked in several ways, but since an eigenvector is needed in the event that one exists, the best strategy is to row reduce the augmented matrix for $(A - 4I)\mathbf{x} = \mathbf{0}$:

$$\begin{bmatrix} -1 & 0 & -1 & 0 \\ 2 & -1 & 1 & 0 \\ -3 & 4 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 4 & 4 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The equation $(A - 4I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, so 4 is an eigenvalue. Any nonzero solution of $(A - 4I)\mathbf{x} = \mathbf{0}$ is a corresponding eigenvector. The entries in a solution satisfy $x_1 + x_3 = 0$ and $-x_2 - x_3 = 0$, with x_3 free. The general solution is *not* requested, so to save time, simply take any nonzero value for x_3 to produce an eigenvector. If $x_3 = 1$, then $\mathbf{x} = (-1, -1, 1)$.

8. To determine if 3 is an eigenvalue of A , decide if the matrix $A - 3I$ is invertible.

$$A - 3I = \begin{bmatrix} 1 & 2 & 2 \\ 3 & -2 & 1 \\ 0 & 1 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 2 & 2 \\ 3 & -5 & 1 \\ 0 & 1 & -2 \end{bmatrix}$$

Row reducing the augmented matrix $[(A - 3I) \ \mathbf{0}]$ yields:

$$\begin{bmatrix} -2 & 2 & 2 & 0 \\ 3 & -5 & 1 & 0 \\ 0 & 1 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & -2 & 4 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The equation $(A - 3I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, so 3 is an eigenvalue. Any nonzero solution of $(A - 3I)\mathbf{x} = \mathbf{0}$ is a corresponding eigenvector. The entries in a solution satisfy $x_1 - 3x_3 = 0$ and $x_2 - 2x_3 = 0$, with x_3 free. The general solution is *not* requested, so to save time, simply take any nonzero value for x_3 to produce an eigenvector. If $x_3 = 1$, then $\mathbf{x} = (3, 2, 1)$.

Question 2:

Find a basis for the eigenspace corresponding to each listed eigenvalue

$$\mathbf{9.} \quad A = \begin{bmatrix} 5 & 0 \\ 2 & 1 \end{bmatrix}, \lambda = 1, 5$$

$$\mathbf{10.} \quad A = \begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}, \lambda = 4$$

$$\mathbf{11.} \quad A = \begin{bmatrix} 4 & -2 \\ -3 & 9 \end{bmatrix}, \lambda = 10$$

$$\mathbf{12.} \quad A = \begin{bmatrix} 7 & 4 \\ -3 & -1 \end{bmatrix}, \lambda = 1, 5$$

$$\mathbf{13.} \quad A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \lambda = 1, 2, 3$$

$$\mathbf{14.} \quad A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & -3 & 0 \\ 4 & -13 & 1 \end{bmatrix}, \lambda = -2$$

$$\mathbf{15.} \quad A = \begin{bmatrix} 4 & 2 & 3 \\ -1 & 1 & -3 \\ 2 & 4 & 9 \end{bmatrix}, \lambda = 3$$

$$\mathbf{16.} \quad A = \begin{bmatrix} 3 & 0 & 2 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}, \lambda = 4$$

Question 2: solution

9. For $\lambda=1$: $A-I = \begin{bmatrix} 5 & 0 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 2 & 0 \end{bmatrix}$

The augmented matrix for $(A-I)\mathbf{x} = \mathbf{0}$ is $\begin{bmatrix} 4 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}$. Thus $x_1 = 0$ and x_2 is free. The general solution of $(A-I)\mathbf{x} = \mathbf{0}$ is $x_2\mathbf{e}_2$, where $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and so $\mathbf{e}_2$ is a basis for the eigenspace corresponding to the eigenvalue 1.

For $\lambda=5$: $A-5I = \begin{bmatrix} 5 & 0 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 2 & -4 \end{bmatrix}$

The equation $(A-5I)\mathbf{x} = \mathbf{0}$ leads to $2x_1 - 4x_2 = 0$, so that $x_1 = 2x_2$ and x_2 is free. The general solution is $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. So $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is a basis for the eigenspace.

10. For $\lambda=4$: $A-4I = \begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 6 & -9 \\ 4 & -6 \end{bmatrix}$.

The augmented matrix for $(A-4I)\mathbf{x} = \mathbf{0}$ is $\begin{bmatrix} 6 & -9 & 0 \\ 4 & -6 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -9/6 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Thus $x_1 = (3/2)x_2$ and x_2 is free. The general solution is $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} (3/2)x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 3/2 \\ 1 \end{bmatrix}$. A basis for the eigenspace corresponding to 4 is $\begin{bmatrix} 3/2 \\ 1 \end{bmatrix}$. Another choice is $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$.

11. $A-10I = \begin{bmatrix} 4 & -2 \\ -3 & 9 \end{bmatrix} - \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} = \begin{bmatrix} -6 & -2 \\ -3 & -1 \end{bmatrix}$

The augmented matrix for $(A-10I)\mathbf{x} = \mathbf{0}$ is $\begin{bmatrix} -6 & -2 & 0 \\ -3 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1/3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Thus $x_1 = (-1/3)x_2$ and x_2 is free. The general solution is $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -(1/3)x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -1/3 \\ 1 \end{bmatrix}$. A basis for the eigenspace corresponding to 10 is $\begin{bmatrix} -1/3 \\ 1 \end{bmatrix}$. Another choice is $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$.

12. For $\lambda=1$: $A-I = \begin{bmatrix} 7 & 4 \\ -3 & -1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 4 \\ -3 & -2 \end{bmatrix}$

The augmented matrix for $(A-I)\mathbf{x} = \mathbf{0}$ is $\begin{bmatrix} 6 & 4 & 0 \\ -3 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2/3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Thus $x_1 = (-2/3)x_2$ and x_2 is free. A basis for the eigenspace corresponding to 1 is $\begin{bmatrix} -2/3 \\ 1 \\ 0 \end{bmatrix}$. Another choice is $\begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix}$.

For $\lambda=5$: $A-5I = \begin{bmatrix} 7 & 4 \\ -3 & -1 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ -3 & -6 \end{bmatrix}$.

13. For $\lambda=1$:

$$A-I = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix}$$

The equations for $(A-I)\mathbf{x} = \mathbf{0}$ are easy to solve: $\begin{cases} 3x_1 + x_3 = 0 \\ -2x_1 = 0 \end{cases}$

Row operations hardly seem necessary. Obviously x_1 is zero, and hence x_3 is also zero. There are three-variables, so x_2 is free. The general solution of $(A-I)\mathbf{x} = \mathbf{0}$ is $x_2\mathbf{e}_2$, where $\mathbf{e}_2 = (0,1,0)$, and so $\mathbf{e}_2$ provides a basis for the eigenspace.

For $\lambda=2$:

$$A-2I = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$[(A-2I) \ \mathbf{0}] = \begin{bmatrix} 2 & 0 & 1 & 0 \\ -2 & -1 & 0 & 0 \\ -2 & 0 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & 0 & 1/2 & 0 \\ 0 & \textcircled{1} & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

So $x_1 = -(1/2)x_3$, $x_2 = x_3$, with x_3 free. The general solution of $(A-2I)\mathbf{x} = \mathbf{0}$ is $x_3 \begin{bmatrix} -1/2 \\ 1 \\ 1 \end{bmatrix}$. A nice basis

vector for the eigenspace is $\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$.

For $\lambda=3$:

$$A-3I = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix}$$

$$[(A-3I) \ \mathbf{0}] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -2 & -2 & 0 & 0 \\ -2 & 0 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & 0 & 1 & 0 \\ 0 & \textcircled{1} & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

So $x_1 = -x_3$, $x_2 = x_3$, with x_3 free. A basis vector for the eigenspace is $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$.

14. For $\lambda = -2$: $A - (-2I) = A + 2I = \begin{bmatrix} 1 & 0 & -1 \\ 1 & -3 & 0 \\ 4 & -13 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & -1 \\ 1 & -1 & 0 \\ 4 & -13 & 3 \end{bmatrix}.$

The augmented matrix for $[A - (-2)I]\mathbf{x} = \mathbf{0}$, or $(A + 2I)\mathbf{x} = \mathbf{0}$, is

$$[(A + 2I) \quad \mathbf{0}] = \begin{bmatrix} 3 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 4 & -13 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1/3 & 0 \\ 0 & 1 & -1/3 & 0 \\ 0 & -13 & 13/3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1/3 & 0 \\ 0 & 1 & -1/3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Thus $x_1 = (1/3)x_3, x_2 = (1/3)x_3$, with x_3 free. The general solution of $(A + 2I)\mathbf{x} = \mathbf{0}$ is $x_3 \begin{bmatrix} 1/3 \\ 1/3 \\ 1 \end{bmatrix}.$

A basis for the eigenspace corresponding to -2 is $\begin{bmatrix} 1/3 \\ 1/3 \\ 1 \end{bmatrix}$; another is $\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}.$

15. For $\lambda = 3$: $[(A - 3I) \quad \mathbf{0}] = \begin{bmatrix} 1 & 2 & 3 & 0 \\ -1 & -2 & -3 & 0 \\ 2 & 4 & 6 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$ Thus $x_1 + 2x_2 + 3x_3 = 0$, with x_2 and

x_3 free. The general solution of $(A - 3I)\mathbf{x} = \mathbf{0}$, is

$$\mathbf{x} = \begin{bmatrix} -2x_2 - 3x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}. \text{Basis for the eigenspace: } \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

16. For $\lambda = 4$: $A - 4I = \begin{bmatrix} 3 & 0 & 2 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 2 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$

$$[(A - 4I) \quad \mathbf{0}] = \begin{bmatrix} -1 & 0 & 2 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \text{ So } x_1 = 2x_3, x_2 = 3x_3, \text{ with } x_3 \text{ and } x_4$$

free variables. The general solution of $(A - 4I)\mathbf{x} = \mathbf{0}$ is

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2x_3 \\ 3x_3 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}. \text{Basis for the eigenspace: } \left\{ \begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Question 3:

Find the eigenvalues of the matrices in Exercises 17 and 18.

17. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 5 \\ 0 & 0 & -1 \end{bmatrix}$

18. $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & -3 \end{bmatrix}$

Question 3: solution

17. The eigenvalues of $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 5 \\ 0 & 0 & -1 \end{bmatrix}$ are 0, 2, and -1 , on the main diagonal, by Theorem 1.

18. The eigenvalues of $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & -3 \end{bmatrix}$ are 4, 0, and -3 , on the main diagonal, by Theorem 1.

Question 4:

Find the characteristic polynomial and the eigenvalues of the matrices in Exercises 1–8.

1. $\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$

2. $\begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$

3. $\begin{bmatrix} 3 & -2 \\ 1 & -1 \end{bmatrix}$

4. $\begin{bmatrix} 5 & -3 \\ -4 & 3 \end{bmatrix}$

5. $\begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$

6. $\begin{bmatrix} 3 & -4 \\ 4 & 8 \end{bmatrix}$

7. $\begin{bmatrix} 5 & 3 \\ -4 & 4 \end{bmatrix}$

8. $\begin{bmatrix} 7 & -2 \\ 2 & 3 \end{bmatrix}$

Question 4: (solution)

1. $A = \begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$, $A - \lambda I = \begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & 7 \\ 7 & 2-\lambda \end{bmatrix}$. The characteristic polynomial is

$$\det(A - \lambda I) = (2 - \lambda)^2 - 7^2 = 4 - 4\lambda + \lambda^2 - 49 = \lambda^2 - 4\lambda - 45$$

In factored form, the characteristic equation is $(\lambda - 9)(\lambda + 5) = 0$, so the eigenvalues of A are 9 and -5 .

2. $A = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$, $A - \lambda I = \begin{bmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{bmatrix}$. The characteristic polynomial is

$$\det(A - \lambda I) = (5 - \lambda)(5 - \lambda) - 3 \cdot 3 = \lambda^2 - 10\lambda + 16$$

Since $\lambda^2 - 10\lambda + 16 = (\lambda - 8)(\lambda - 2)$, the eigenvalues of A are 8 and 2.

3. $A = \begin{bmatrix} 3 & -2 \\ 1 & -1 \end{bmatrix}$, $A - \lambda I = \begin{bmatrix} 3-\lambda & -2 \\ 1 & -1-\lambda \end{bmatrix}$. The characteristic polynomial is

$$\det(A - \lambda I) = (3 - \lambda)(-1 - \lambda) - (-2)(1) = \lambda^2 - 2\lambda - 1$$

Use the quadratic formula to solve the characteristic equation and find the eigenvalues:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{2 \pm \sqrt{4 + 4}}{2} = 1 \pm \sqrt{2}$$

4. $A = \begin{bmatrix} 5 & -3 \\ -4 & 3 \end{bmatrix}$, $A - \lambda I = \begin{bmatrix} 5-\lambda & -3 \\ -4 & 3-\lambda \end{bmatrix}$. The characteristic polynomial of A is

$$\det(A - \lambda I) = (5 - \lambda)(3 - \lambda) - (-3)(-4) = \lambda^2 - 8\lambda + 3$$

Use the quadratic formula to solve the characteristic equation and find the eigenvalues:

$$\lambda = \frac{8 \pm \sqrt{64 - 4(3)}}{2} = \frac{8 \pm 2\sqrt{13}}{2} = 4 \pm \sqrt{13}$$

5. $A = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$, $A - \lambda I = \begin{bmatrix} 2-\lambda & 1 \\ -1 & 4-\lambda \end{bmatrix}$. The characteristic polynomial of A is

$$\det(A - \lambda I) = (2 - \lambda)(4 - \lambda) - (1)(-1) = \lambda^2 - 6\lambda + 9 = (\lambda - 3)^2$$

Thus, A has only one eigenvalue 3, with multiplicity 2.

6. $A = \begin{bmatrix} 3 & -4 \\ 4 & 8 \end{bmatrix}$, $A - \lambda I = \begin{bmatrix} 3-\lambda & -4 \\ 4 & 8-\lambda \end{bmatrix}$. The characteristic polynomial is

$$\det(A - \lambda I) = (3 - \lambda)(8 - \lambda) - (-4)(4) = \lambda^2 - 11\lambda + 40$$

Use the quadratic formula to solve $\det (A - \lambda I) = 0$:

$$\lambda = \frac{-11 \pm \sqrt{121 - 4(40)}}{2} = \frac{-11 \pm \sqrt{-39}}{2}$$

These values are complex numbers, not real numbers, so A has no real eigenvalues. There is no nonzero vector $\mathbf{x}$ in $\mathbf{R}^2$ such that $A\mathbf{x} = \lambda\mathbf{x}$, because a real vector $A\mathbf{x}$ cannot equal a complex multiple of $\mathbf{x}$.

7. $A = \begin{bmatrix} 5 & 3 \\ -4 & 4 \end{bmatrix}$, $A - \lambda I = \begin{bmatrix} 5 - \lambda & 3 \\ -4 & 4 - \lambda \end{bmatrix}$. The characteristic polynomial is

$$\det(A - \lambda I) = (5 - \lambda)(4 - \lambda) - (3)(-4) = \lambda^2 - 9\lambda + 32$$

Use the quadratic formula to solve $\det (A - \lambda I) = 0$:

$$\lambda = \frac{9 \pm \sqrt{81 - 4(32)}}{2} = \frac{9 \pm \sqrt{-47}}{2}$$

These values are complex numbers, not real numbers, so A has no real eigenvalues. There is no nonzero vector $\mathbf{x}$ in $\mathbf{R}^2$ such that $A\mathbf{x} = \lambda\mathbf{x}$, because a real vector $A\mathbf{x}$ cannot equal a complex multiple of $\mathbf{x}$.

8. $A = \begin{bmatrix} 7 & -2 \\ 2 & 3 \end{bmatrix}$, $A - \lambda I = \begin{bmatrix} 7 - \lambda & -2 \\ 2 & 3 - \lambda \end{bmatrix}$. The characteristic polynomial is

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) - (-2)(2) = \lambda^2 - 10\lambda + 25$$

Since $\lambda^2 - 10\lambda + 25 = (\lambda - 5)^2$, the only eigenvalue is 5, with multiplicity 2.

Question 5:

Find the characteristic polynomial of the following matrices:

9. $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & -1 \\ 0 & 6 & 0 \end{bmatrix}$

10. $\begin{bmatrix} 0 & 3 & 1 \\ 3 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$

11. $\begin{bmatrix} 4 & 0 & 0 \\ 5 & 3 & 2 \\ -2 & 0 & 2 \end{bmatrix}$

12. $\begin{bmatrix} -1 & 0 & 1 \\ -3 & 4 & 1 \\ 0 & 0 & 2 \end{bmatrix}$

13. $\begin{bmatrix} 6 & -2 & 0 \\ -2 & 9 & 0 \\ 5 & 8 & 3 \end{bmatrix}$

14. $\begin{bmatrix} 5 & -2 & 3 \\ 0 & 1 & 0 \\ 6 & 7 & -2 \end{bmatrix}$

Question 5: solution

9. $\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 0 & -1 \\ 2 & 3-\lambda & -1 \\ 0 & 6 & 0-\lambda \end{bmatrix}$. From the special formula for 3×3 determinants, the

characteristic polynomial is

$$\begin{aligned} \det(A - \lambda I) &= (1-\lambda)(3-\lambda)(-\lambda) + 0 + (-1)(2)(6) - 0 - (6)(-1)(1-\lambda) - 0 \\ &= (\lambda^2 - 4\lambda + 3)(-\lambda) - 12 + 6(1-\lambda) \\ &= -\lambda^3 + 4\lambda^2 - 3\lambda - 12 + 6 - 6\lambda \\ &= -\lambda^3 + 4\lambda^2 - 9\lambda - 6 \end{aligned}$$

(This polynomial has one irrational zero and two imaginary zeros.) Another way to evaluate the determinant is to interchange rows 1 and 2 (which reverses the sign of the determinant) and then make one row replacement:

$$\begin{aligned} \det \begin{bmatrix} 1-\lambda & 0 & -1 \\ 2 & 3-\lambda & -1 \\ 0 & 6 & 0-\lambda \end{bmatrix} &= -\det \begin{bmatrix} 2 & 3-\lambda & -1 \\ 1-\lambda & 0 & -1 \\ 0 & 6 & 0-\lambda \end{bmatrix} \\ &= -\det \begin{bmatrix} 2 & 3-\lambda & -1 \\ 0 & 0 + (.5\lambda - .5)(3-\lambda) & -1 + (.5\lambda - .5)(-1) \\ 0 & 6 & 0-\lambda \end{bmatrix} \end{aligned}$$

Next, expand by cofactors down the first column. The quantity above equals

$$\begin{aligned} -2 \det \begin{bmatrix} (.5\lambda - .5)(3-\lambda) & -.5 - .5\lambda \\ 6 & -\lambda \end{bmatrix} &= -2[(.5\lambda - .5)(3-\lambda)(-\lambda) - (-.5 - .5\lambda)(6)] \\ &= (1-\lambda)(3-\lambda)(-\lambda) - (1+\lambda)(6) = (\lambda^2 - 4\lambda + 3)(-\lambda) - 6 - 6\lambda = -\lambda^3 + 4\lambda^2 - 9\lambda - 6 \end{aligned}$$

10. $\det(A - \lambda I) = \det \begin{bmatrix} 0-\lambda & 3 & 1 \\ 3 & 0-\lambda & 2 \\ 1 & 2 & 0-\lambda \end{bmatrix}$. From the special formula for 3×3 determinants, the

characteristic polynomial is

$$\begin{aligned} \det(A - \lambda I) &= (-\lambda)(-\lambda)(-\lambda) + 3 \cdot 2 \cdot 1 + 1 \cdot 3 \cdot 2 - 1 \cdot (-\lambda) \cdot 1 - 2 \cdot 2 \cdot (-\lambda) - (-\lambda) \cdot 3 \cdot 3 \\ &= -\lambda^3 + 6 + 6 + \lambda + 4\lambda + 9\lambda = -\lambda^3 + 14\lambda + 12 \end{aligned}$$

11. The special arrangements of zeros in A makes a cofactor expansion along the first row highly effective.

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} 4-\lambda & 0 & 0 \\ 5 & 3-\lambda & 2 \\ -2 & 0 & 2-\lambda \end{bmatrix} = (4-\lambda) \det \begin{bmatrix} 3-\lambda & 2 \\ 0 & 2-\lambda \end{bmatrix} \\ &= (4-\lambda)(3-\lambda)(2-\lambda) = (4-\lambda)(\lambda^2 - 5\lambda + 6) = -\lambda^3 + 9\lambda^2 - 26\lambda + 24 \end{aligned}$$

If only the eigenvalues were required, there would be no need here to write the characteristic polynomial in expanded form.

12. Make a cofactor expansion along the third row:

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} -1-\lambda & 0 & 1 \\ -3 & 4-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{bmatrix} = (2-\lambda) \cdot \det \begin{bmatrix} -1-\lambda & 0 \\ -3 & 4-\lambda \end{bmatrix} \\ &= (2-\lambda)(-1-\lambda)(4-\lambda) = -\lambda^3 + 5\lambda^2 - 2\lambda - 8 \end{aligned}$$

13. Make a cofactor expansion down the third column:

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} 6-\lambda & -2 & 0 \\ -2 & 9-\lambda & 0 \\ 5 & 8 & 3-\lambda \end{bmatrix} = (3-\lambda) \cdot \det \begin{bmatrix} 6-\lambda & -2 \\ -2 & 9-\lambda \end{bmatrix} \\ &= (3-\lambda)[(6-\lambda)(9-\lambda) - (-2)(-2)] = (3-\lambda)(\lambda^2 - 15\lambda + 50) \\ &= -\lambda^3 + 18\lambda^2 - 95\lambda + 150 \text{ or } (3-\lambda)(\lambda-5)(\lambda-10) \end{aligned}$$

14. Make a cofactor expansion along the second row:

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} 5-\lambda & -2 & 3 \\ 0 & 1-\lambda & 0 \\ 6 & 7 & -2-\lambda \end{bmatrix} = (1-\lambda) \cdot \det \begin{bmatrix} 5-\lambda & 3 \\ 6 & -2-\lambda \end{bmatrix} \\ &= (1-\lambda) \cdot [(5-\lambda)(-2-\lambda) - 3 \cdot 6] = (1-\lambda)(\lambda^2 - 3\lambda - 28) \\ &= -\lambda^3 + 4\lambda^2 + 25\lambda - 28 \text{ or } (1-\lambda)(\lambda-7)(\lambda+4) \end{aligned}$$

Question 6:

For the matrices in Exercises 15–17, list the eigenvalues, repeated according to their multiplicities.

$$15. \begin{bmatrix} 4 & -7 & 0 & 2 \\ 0 & 3 & -4 & 6 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$16. \begin{bmatrix} 5 & 0 & 0 & 0 \\ 8 & -4 & 0 & 0 \\ 0 & 7 & 1 & 0 \\ 1 & -5 & 2 & 1 \end{bmatrix}$$

$$17. \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ -5 & 1 & 0 & 0 & 0 \\ 3 & 8 & 0 & 0 & 0 \\ 0 & -7 & 2 & 1 & 0 \\ -4 & 1 & 9 & -2 & 3 \end{bmatrix}$$

Question 6: solution

15. Use the fact that the determinant of a triangular matrix is the product of the diagonal entries:

$$\det(A - \lambda I) = \det \begin{bmatrix} 4 - \lambda & -7 & 0 & 2 \\ 0 & 3 - \lambda & -4 & 6 \\ 0 & 0 & 3 - \lambda & -8 \\ 0 & 0 & 0 & 1 - \lambda \end{bmatrix} = (4 - \lambda)(3 - \lambda)^2(1 - \lambda)$$

The eigenvalues are 4, 3, 3, and 1.

16. The determinant of a triangular matrix is the product of its diagonal entries:

$$\det(A - \lambda I) = \det \begin{bmatrix} 5 - \lambda & 0 & 0 & 0 \\ 8 & -4 - \lambda & 0 & 0 \\ 0 & 7 & 1 - \lambda & 0 \\ 1 & -5 & 2 & 1 - \lambda \end{bmatrix} = (5 - \lambda)(-4 - \lambda)(1 - \lambda)^2$$

The eigenvalues are 5, 1, 1, and -4.

17. The determinant of a triangular matrix is the product of its diagonal entries:

$$\begin{bmatrix} 3 - \lambda & 0 & 0 & 0 & 0 \\ -5 & 1 - \lambda & 0 & 0 & 0 \\ 3 & 8 & 0 - \lambda & 0 & 0 \\ 0 & -7 & 2 & 1 - \lambda & 0 \\ -4 & 1 & 9 & -2 & 3 - \lambda \end{bmatrix} = (3 - \lambda)^2(1 - \lambda)^2(-\lambda)$$

The eigenvalues are 3, 3, 1, 1, and 0.

Question 7:

In Exercises 1 and 2, let $A = PDP^{-1}$ and compute A^4 .

$$1. \quad P = \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix}, D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$2. \quad P = \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$$

Question 7: solution

$$1. P = \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix}, D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, A = PDP^{-1}, \text{ and } A^4 = PD^4P^{-1}. \text{ We compute } P^{-1} = \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix}, D^4 = \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix},$$

$$\text{and } A^4 = \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 226 & -525 \\ 90 & -209 \end{bmatrix}$$

$$2. P = \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}, A = PDP^{-1}, \text{ and } A^4 = PD^4P^{-1}. \text{ We compute}$$

$$P^{-1} = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}, D^4 = \begin{bmatrix} 1 & 0 \\ 0 & 1/16 \end{bmatrix}, \text{ and } A^4 = \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1/16 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix} = \frac{1}{16} \begin{bmatrix} 151 & 90 \\ -225 & -134 \end{bmatrix}$$

Question 8:

In Exercises 5 and 6, the matrix A is factored in the form PDP^{-1} . Use the Diagonalization Theorem to find the eigenvalues of A and a basis for each eigenspace.

$$5. \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/4 & 1/2 & -3/4 \\ 1/4 & -1/2 & 1/4 \end{bmatrix}$$

$$6. \begin{bmatrix} 4 & 0 & -2 \\ 2 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix} =$$

$$\begin{bmatrix} -2 & 0 & -1 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 4 \\ -1 & 0 & -2 \end{bmatrix}$$

Question 8: solution

5. By the Diagonalization Theorem, eigenvectors form the columns of the left factor, and they correspond respectively to the eigenvalues on the diagonal of the middle factor.

$$\lambda = 5: \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}; \lambda = 1: \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$$

6. As in Exercise 5, inspection of the factorization gives:

$$\lambda = 4: \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}; \lambda = 5: \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Question 9:

$$7. \begin{bmatrix} 1 & 0 \\ 6 & -1 \end{bmatrix}$$

$$8. \begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix}$$

$$9. \begin{bmatrix} 3 & -1 \\ 1 & 5 \end{bmatrix}$$

$$10. \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$$

$$11. \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$$

$$12. \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$$

$$13. \begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{bmatrix}$$

$$14. \begin{bmatrix} 4 & 0 & -2 \\ 2 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix}$$

Question 9: solution

7. Since A is triangular, its eigenvalues are obviously ± 1 .

For $\lambda = 1$: $A - I = \begin{bmatrix} 0 & 0 \\ 6 & -2 \end{bmatrix}$. The equation $(A - I)\mathbf{x} = \mathbf{0}$ amounts to $6x_1 - 2x_2 = 0$, so $x_1 = (1/3)x_2$ with x_2 free. The general solution is $x_2 \begin{bmatrix} 1/3 \\ 1 \end{bmatrix}$, and a nice basis vector for the eigenspace is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$.

For $\lambda = -1$: $A + I = \begin{bmatrix} 2 & 0 \\ 6 & 0 \end{bmatrix}$. The equation $(A + I)\mathbf{x} = \mathbf{0}$ amounts to $2x_1 = 0$, so $x_1 = 0$ with x_2 free. The general solution is $x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and a basis vector for the eigenspace is $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

From $\mathbf{v}_1$ and $\mathbf{v}_2$ construct $P = [\mathbf{v}_1 \quad \mathbf{v}_2] = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$. Then set $D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, where the eigenvalues in D correspond to $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively.

8. Since A is triangular, its only eigenvalue is obviously 5.

For $\lambda = 5$: $A - 5I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. The equation $(A - 5I)\mathbf{x} = \mathbf{0}$ amounts to $x_2 = 0$, so $x_2 = 0$ with x_1 free. The general solution is $x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Since we cannot generate an eigenvector basis for $\mathbb{R}^2$, A is not diagonalizable.

9. To find the eigenvalues of A , compute its characteristic polynomial:

$$\det(A - \lambda I) = \det \begin{bmatrix} 3 - \lambda & -1 \\ 1 & 5 - \lambda \end{bmatrix} = (3 - \lambda)(5 - \lambda) - (-1)(1) = \lambda^2 - 8\lambda + 16 = (\lambda - 4)^2$$

Thus the only eigenvalue of A is 4.

For $\lambda = 4$: $A - 4I = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$. The equation $(A - 4I)\mathbf{x} = \mathbf{0}$ amounts to $x_1 + x_2 = 0$, so $x_1 = -x_2$ with x_2 free. The general solution is $x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Since we cannot generate an eigenvector basis for $\mathbb{R}^2$, A is not diagonalizable.

10. To find the eigenvalues of A , compute its characteristic polynomial:

$$\det(A - \lambda I) = \det \begin{bmatrix} 2 - \lambda & 3 \\ 4 & 1 - \lambda \end{bmatrix} = (2 - \lambda)(1 - \lambda) - (3)(4) = \lambda^2 - 3\lambda - 10 = (\lambda - 5)(\lambda + 2)$$

Thus the eigenvalues of A are 5 and -2 .

For $\lambda = 5$: $A - 5I = \begin{bmatrix} -3 & 3 \\ 4 & -4 \end{bmatrix}$. The equation $(A - 5I)\mathbf{x} = \mathbf{0}$ amounts to $x_1 - x_2 = 0$, so $x_1 = x_2$ with x_2 free. The general solution is $x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and a basis vector for the eigenspace is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

For $\lambda = -2$: $A + 2I = \begin{bmatrix} 4 & 3 \\ 4 & 3 \end{bmatrix}$. The equation $(A + 2I)\mathbf{x} = \mathbf{0}$ amounts to $4x_1 + 3x_2 = 0$, so $x_1 = (-3/4)x_2$ with x_2 free. The general solution is $x_2 \begin{bmatrix} -3/4 \\ 1 \end{bmatrix}$, and a nice basis vector for the eigenspace is $\mathbf{v}_2 = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$.

From $\mathbf{v}_1$ and $\mathbf{v}_2$ construct $P = [\mathbf{v}_1 \quad \mathbf{v}_2] = \begin{bmatrix} 1 & -3 \\ 1 & 4 \end{bmatrix}$. Then set $D = \begin{bmatrix} 5 & 0 \\ 0 & -2 \end{bmatrix}$, where the eigenvalues in D correspond to $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively.

11. The eigenvalues of A are given to be 1, 2, and 3.

For $\lambda = 3$: $A - 3I = \begin{bmatrix} -4 & 4 & -2 \\ -3 & 1 & 0 \\ -3 & 1 & 0 \end{bmatrix}$, and row reducing $[A - 3I \quad \mathbf{0}]$ yields $\begin{bmatrix} 1 & 0 & -1/4 & 0 \\ 0 & 1 & -3/4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The general solution is $x_3 \begin{bmatrix} 1/4 \\ 3/4 \\ 1 \end{bmatrix}$, and a nice basis vector for the eigenspace is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$.

For $\lambda=2$: $A-2I = \begin{bmatrix} -3 & 4 & -2 \\ -3 & 2 & 0 \\ -3 & 1 & 1 \end{bmatrix}$, and row reducing $[A-2I \quad \mathbf{0}]$ yields $\begin{bmatrix} 1 & 0 & -2/3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The

general solution is $x_3 \begin{bmatrix} 2/3 \\ 1 \\ 1 \end{bmatrix}$, and a nice basis vector for the eigenspace is $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$.

For $\lambda=1$: $A-I = \begin{bmatrix} -2 & 4 & -2 \\ -3 & 3 & 0 \\ -3 & 1 & 2 \end{bmatrix}$, and row reducing $[A-I \quad \mathbf{0}]$ yields $\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The general

solution is $x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and a basis vector for the eigenspace is $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

From $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ construct $P = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 3 & 1 \\ 4 & 3 & 1 \end{bmatrix}$. Then set $D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, where the

eigenvalues in D correspond to $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ respectively.

12. The eigenvalues of A are given to be 2 and 8.

For $\lambda=8$: $A-8I = \begin{bmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{bmatrix}$, and row reducing $[A-8I \quad \mathbf{0}]$ yields $\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The

general solution is $x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and a basis vector for the eigenspace is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

For $\lambda=2$: $A-2I = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$, and row reducing $[A-2I \quad \mathbf{0}]$ yields $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The general

solution is $x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, and a basis for the eigenspace is $\{\mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$.

From $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ construct $P = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$. Then set $D = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, where the

eigenvalues in D correspond to $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ respectively.

13. The eigenvalues of A are given to be 5 and 1.

For $\lambda = 5$: $A - 5I = \begin{bmatrix} -3 & 2 & -1 \\ 1 & -2 & -1 \\ -1 & -2 & -3 \end{bmatrix}$, and row reducing $[A - 5I \quad \mathbf{0}]$ yields $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The general

solution is $x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$, and a basis for the eigenspace is $\mathbf{v}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$.

For $\lambda = 1$: $A - I = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & -1 \\ -1 & -2 & 1 \end{bmatrix}$, and row reducing $[A - I \quad \mathbf{0}]$ yields $\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The general

solution is $x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, and a basis for the eigenspace is $\{\mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$.

From $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ construct $P = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] = \begin{bmatrix} -1 & -2 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$. Then set $D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, where the eigenvalues in D correspond to $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ respectively.

14. The eigenvalues of A are given to be 5 and 4.

For $\lambda = 5$: $A - 5I = \begin{bmatrix} -1 & 0 & -2 \\ 2 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix}$, and row reducing $[A - 5I \quad \mathbf{0}]$ yields $\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The general

solution is $x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$, and a basis for the eigenspace is $\{\mathbf{v}_1, \mathbf{v}_2\} = \left\{ \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$.

For $\lambda = 4$: $A - 4I = \begin{bmatrix} 0 & 0 & -2 \\ 2 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$, and row reducing $[A - 4I \quad \mathbf{0}]$ yields $\begin{bmatrix} 1 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The general

solution is $x_3 \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix}$, and a nice basis vector for the eigenspace is $\mathbf{v}_3 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$.

From $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ construct $P = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] = \begin{bmatrix} -2 & 0 & -1 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{bmatrix}$. Then set $D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 4 \end{bmatrix}$, where the eigenvalues in D correspond to $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ respectively.

Question 10:

1. The vectors in the given sequence approach an eigenvector $\mathbf{v}_1$. The last vector in the sequence,

$\mathbf{x}_4 = \begin{bmatrix} 1 \\ .3326 \end{bmatrix}$, is probably the best estimate for $\mathbf{v}_1$. To compute an estimate for λ_1 , examine

$A\mathbf{x}_4 = \begin{bmatrix} 4.9978 \\ 1.6652 \end{bmatrix}$. This vector is approximately $\lambda_1 \mathbf{v}_1$. From the first entry in this vector, an estimate of λ_1 is 4.9978.

2. The vectors in the given sequence approach an eigenvector $\mathbf{v}_1$. The last vector in the sequence,

$\mathbf{x}_4 = \begin{bmatrix} -.2520 \\ 1 \end{bmatrix}$, is probably the best estimate for $\mathbf{v}_1$. To compute an estimate for λ_1 , examine

$A\mathbf{x}_4 = \begin{bmatrix} -1.2536 \\ 5.0064 \end{bmatrix}$. This vector is approximately $\lambda_1 \mathbf{v}_1$. From the second entry in this vector, an estimate of λ_1 is 5.0064.

3. The vectors in the given sequence approach an eigenvector $\mathbf{v}_1$. The last vector in the sequence,

$\mathbf{x}_4 = \begin{bmatrix} .5188 \\ 1 \end{bmatrix}$, is probably the best estimate for $\mathbf{v}_1$. To compute an estimate for λ_1 , examine

$A\mathbf{x}_4 = \begin{bmatrix} .4594 \\ .9075 \end{bmatrix}$. This vector is approximately $\lambda_1 \mathbf{v}_1$. From the second entry in this vector, an estimate of λ_1 is .9075.

4. The vectors in the given sequence approach an eigenvector $\mathbf{v}_1$. The last vector in the sequence,

$\mathbf{x}_4 = \begin{bmatrix} 1 \\ .7502 \end{bmatrix}$, is probably the best estimate for $\mathbf{v}_1$. To compute an estimate for λ_1 , examine

$A\mathbf{x}_4 = \begin{bmatrix} -.4012 \\ -.3009 \end{bmatrix}$. This vector is approximately $\lambda_1 \mathbf{v}_1$. From the first entry in this vector, an estimate of λ_1 is $-.4012$.

Question 11:

[M] $A = \begin{bmatrix} 6 & 7 \\ 8 & 5 \end{bmatrix}$, $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. The data in the table below was calculated using Mathematica, which carried more digits than shown here.

k	0	1	2	3	4	5
$\mathbf{x}_k$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} .75 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ .9565 \end{bmatrix}$	$\begin{bmatrix} .9932 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ .9990 \end{bmatrix}$	$\begin{bmatrix} .9998 \\ 1 \end{bmatrix}$
$A\mathbf{x}_k$	$\begin{bmatrix} 6 \\ 8 \end{bmatrix}$	$\begin{bmatrix} 11.5 \\ 11.0 \end{bmatrix}$	$\begin{bmatrix} 12.6957 \\ 12.7826 \end{bmatrix}$	$\begin{bmatrix} 12.9592 \\ 12.9456 \end{bmatrix}$	$\begin{bmatrix} 12.9927 \\ 12.9948 \end{bmatrix}$	$\begin{bmatrix} 12.9990 \\ 12.9987 \end{bmatrix}$
μ_k	8	11.5	12.7826	12.9592	12.9948	12.9990

The actual eigenvalue is 13.